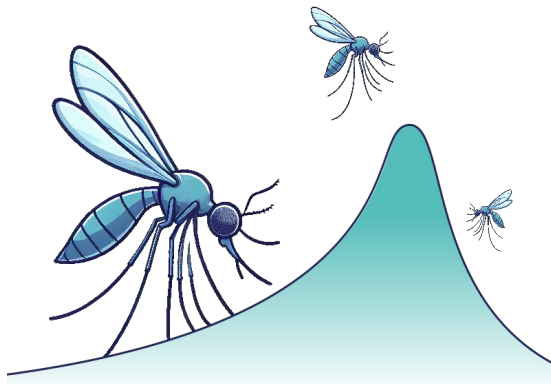




Modelling *Aedes koreicus*: overview of correlative and mechanistic approaches

What we know, what we can do, and what is missing

Daniele Da Re
Giovanni Marini



FONDAZIONE
EDMUND MACH
dal 1874

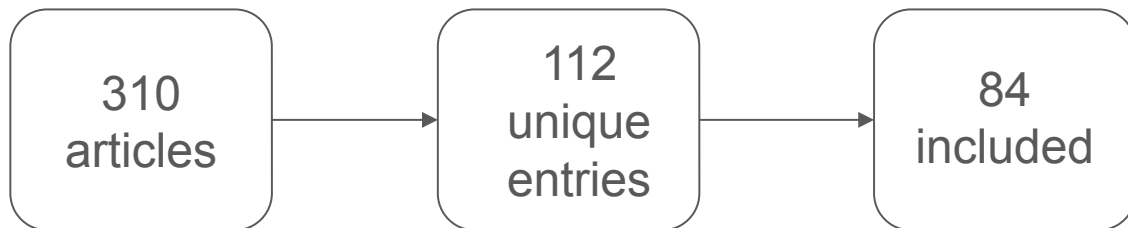


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the European Union

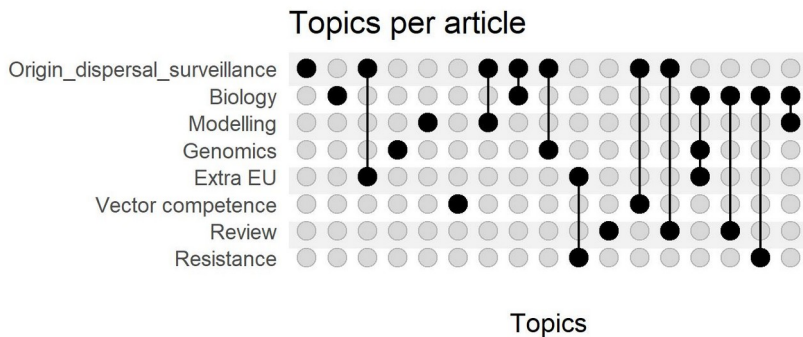
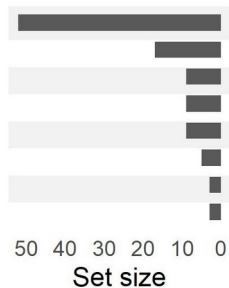
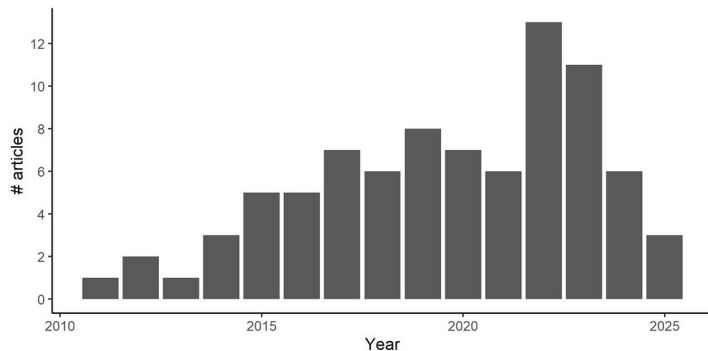
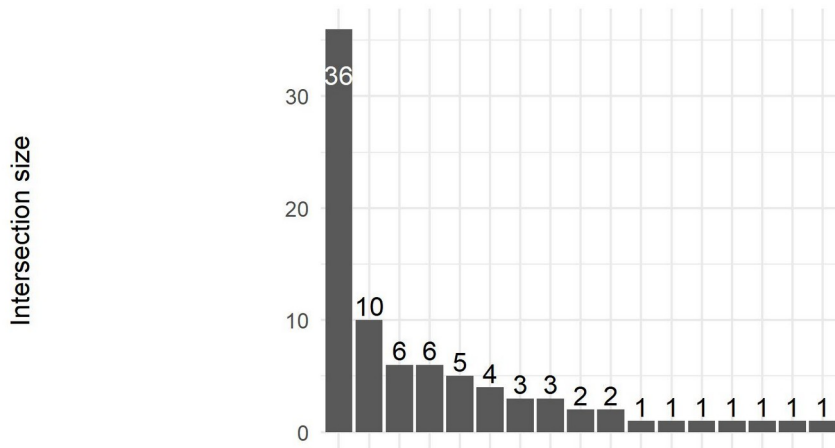
What we know? Literature search

“Aedes koreicus” / “Ae. koreicus” in Scopus and Web of Science (title, abstract, key words)

Search carried out during summer 2025.



What we know? Literature search



Biology

1	MONTARSI, F, 2013
2	DAMIANI, C, 2022
3	BALDACCHINO, F, 2017
4	CIOCCHETTA, S, 2023
5	ROSSO, F, 2018
6	MONTARSI, F, 2022
7	ALFANO, N, 2019
8	BALDACCHINO, F, 2017
9	MARCONDES, CB, 2018
10	SHIN, J, 2020
11	ANICIC, N, 2023
12	NEGRI, A, 2024
13	MARINI, G, 2022
14	SAUER, FG, 2023
15	MARINI, G, 2019
16	CEBRIAN-CAMISON, S, 2020
17	CIOCCHETTA, S, 2017

Vector competence

1	JANSEN, S, 2021
2	KURUCZ, K, 2018
3	CIOCCHETTA, S, 2018
4	MONTARSI, F, 2015
5	JANSEN, S, 2022

Origin, dispersal, surveillance

1	CHERIX, D, 2019
2	LEE, D-Y, 2024
3	MONTARSI, F, 2013
4	SORESINETTI, L, 2025
5	ARNOLDI, I, 2022
6	MOSCA, A, 2022
7	GRADONI, F, 2021
8	STEINBRINK, A, 2019
9	SUTER, T, 2015
10	KALAN, K, 2017
11	KURUCZ, K, 2020
12	KURUCZ, K, 2018
13	VERSTEIRT, V, 2015
14	SCHNEIDER, J, 2016
15	VERSTEIRT, V, 2012
16	BALDACCHINO, F, 2017
17	WERNER, D, 2020
18	ANDREEVA, YV, 2021
19	GANUSHKINA, LA, 2016
20	BEZZHONOVA, OV, 2014
21	WALTHER, D, 2017
22	KURUCZ, K, 2016
23	WERNER, D, 2016
24	KAMPEN, H, 2017
25	ANICIC, N, 2023
26	FUEHRER, H-P, 2020
27	SEOK, S, 2024
28	KURUCZ, K, 2022
29	VOJTISEK, J, 2022
30	GANUSHKINA, L, 2020
31	VERSTEIRT, V, 2012
32	DESIATO, R, 2021
33	NEGRI, A, 2021
34	SAUER, FG, 2023
35	MONTARSI, F, 2015
36	HOHMEISTER, N, 2021
37	CULL, B, 2025
38	SCHENKEL, CD, 2019
39	PFITZNER, WP, 2018
40	BALLARDINI, M, 2019
41	DEBLAUWE, I, 2022
42	ZAMBURLINI, R, 2019
43	GARAMSZEGLI, LZ, 2023
44	CAPELLI, G, 2011
45	MIRANDA, MA, 2022
46	BOUKRAA, S, 2015
47	JUZNIC-ZONTA, Z, 2022
48	ROMI, R, 2018
49	DEBLAUWE, I, 2014
50	MULLER, P, 2020
51	WITTWER, C, 2024
52	GARAMSZEGLI, LZ, 2023

Review

1	GANASSI, S, 2022
2	CEBRIAN-CAMISON, S, 2020
3	ROMI, R, 2018

Genomics

1	CATAPANO, PL, 2023
2	SORESINETTI, L, 2025
3	ZADRA, N, 2023
4	NAGY, NA, 2024
5	SHIN, J, 2020
6	SCHAFFNER, F, 2014
7	KURUCZ, K, 2022
8	SORESINETTI, L, 2023
9	REICHL, J, 2024

Resistance

1	RYU, J, 2019
2	STOOPS, CA, 2023
3	NEGRI, A, 2024

Modelling

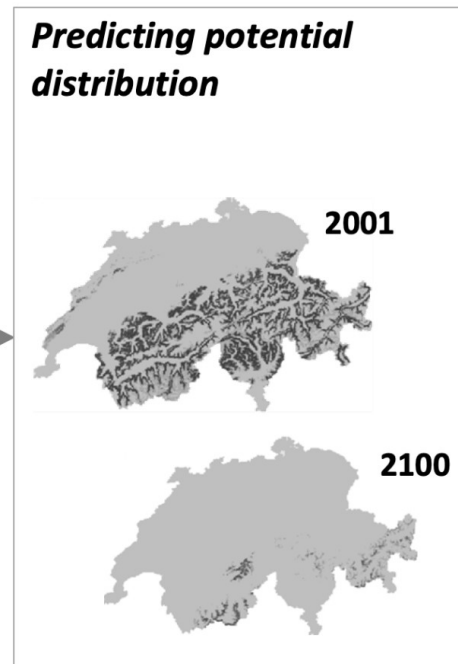
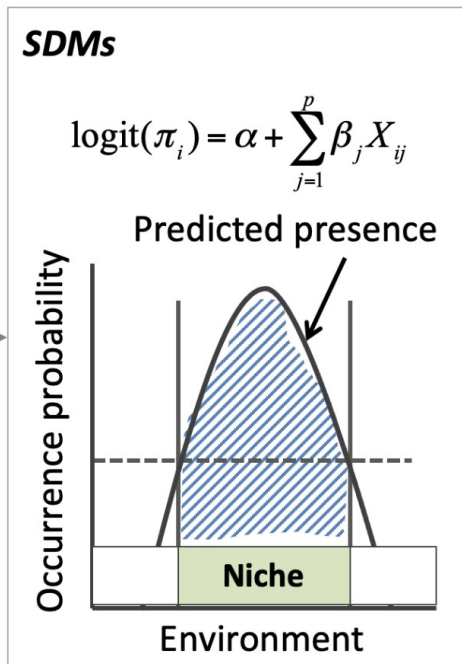
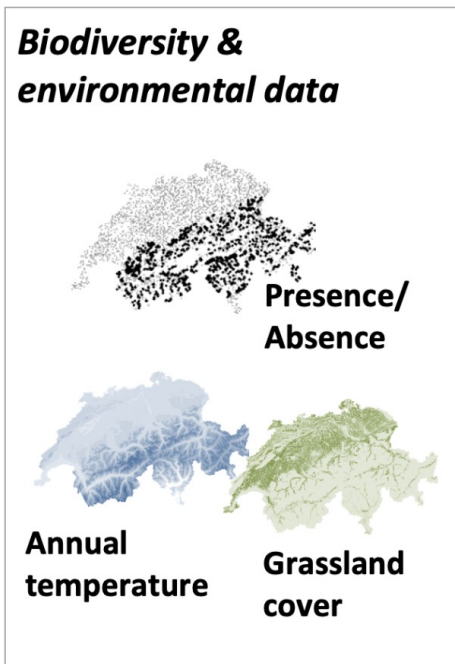
1	MARCANTONIO, M, 2016
2	ARNOLDI, I, 2022
3	KURUCZ, K, 2020
4	LIU, Q, 2023
5	SEOK, S, 2023
6	MARINI, G, 2019
7	GARAMSZEGLI, LZ, 2023
8	DA RE, D, 2022
9	GARAMSZEGLI, LZ, 2025

Extra EU

1	LEE, D-Y, 2024
2	RYU, J, 2019
3	ANDREEVA, YV, 2021
4	GANUSHKINA, LA, 2016
5	BEZZHONOVA, OV, 2014
6	SHIN, J, 2020
7	STOOPS, CA, 2023
8	SEOK, S, 2024
9	GANUSHKINA, L, 2020



What we can do? Correlative models





First assessment of potential distribution and dispersal capacity of the emerging invasive mosquito *Aedes koreicus* in Northeast Italy

Matteo Marcantonio^{1*}, Markus Metz¹, Frédéric Baldacchino¹, Daniele Arnoldi¹, Fabrizio Montarsi², Gioia Capelli², Sara Carlin², Markus Neteler¹ and Annapaola Rizzoli¹

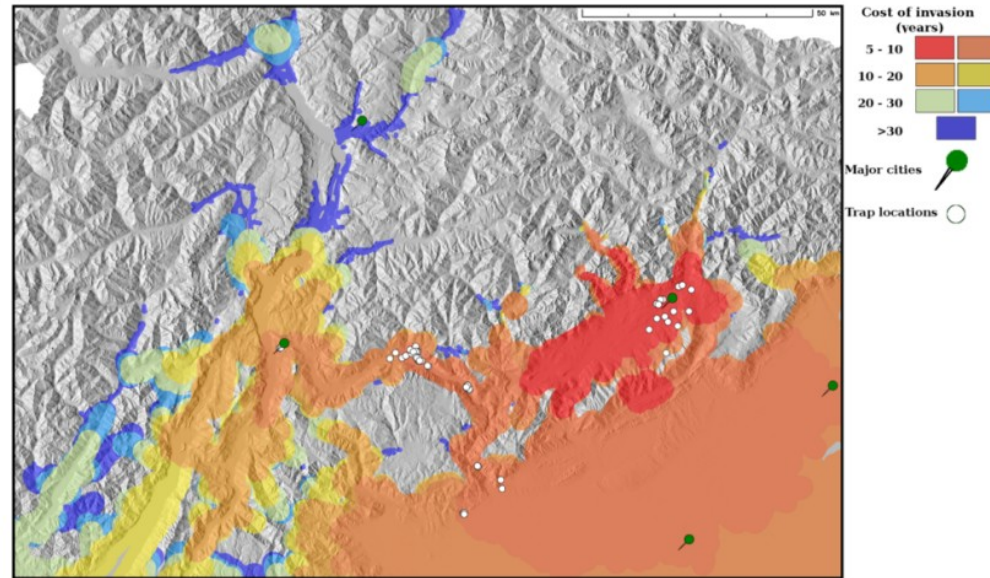


Fig. 4 Potential spread of *Aedes koreicus* predicted through road network analysis: Areas with the same cost of invasion are displayed using a red-green-blue colour scale. The cost of invasion is expressed in years since the species' introduction (2011). Cost of invasion is a function of the travelling distance from the introduction point based on the observed rate of shift of the invaded range centroid and the predicted habitat suitability. Major cities (green pushpins) and sampling locations (white circles) are also reported



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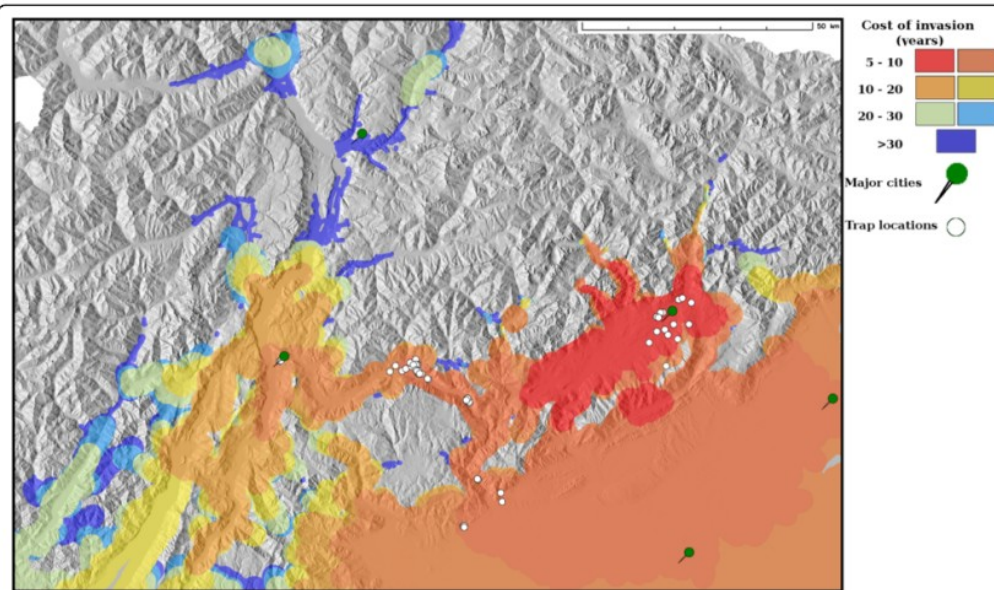
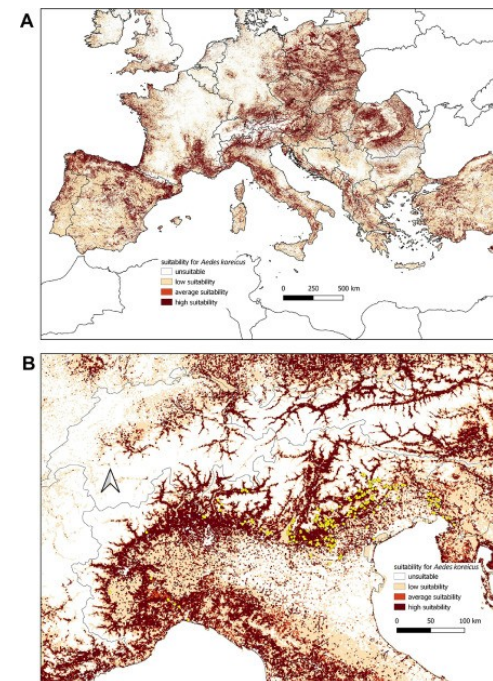


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Assessing the distribution of invasive Asian mosquitoes in Northern Italy and modelling the potential spread of *Aedes koreicus* in Europe

Irene Arnoldi^{a b c d}, Agata Negri^{a b}, Laura Soresinetti^{a c}, Mattia Brambilla^e, Davide Carraretto^{c d}, Fabrizio Montarsi^f, Paolo Roberto^g, Andrea Mosca^g, Diego Rubolini^e, Claudio Bandi^{a b}, Sara Epis^{a b} , Paolo Gabrieli^{a b}





First assessment of potential distribution and dispersal capacity of the emerging invasive mosquito *Aedes koreicus* in Northeast Italy



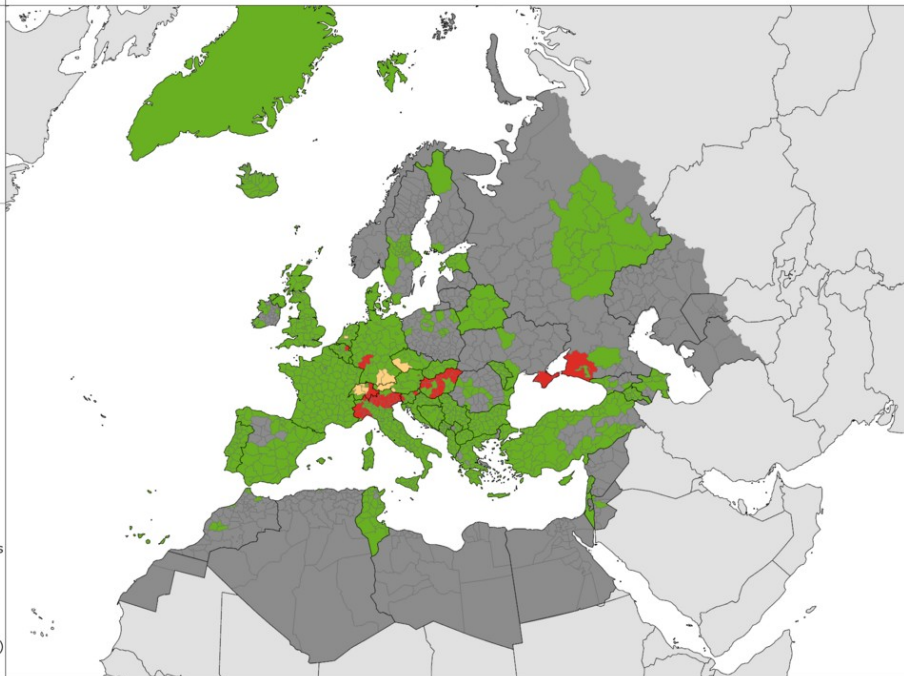
Aedes koreicus, October 2023

Legend

- Established
- Introduced
- Absent
- No data
- Unknown

Countries/Regions not viewable in the main map extent*

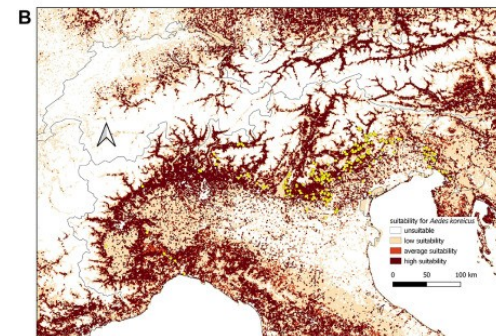
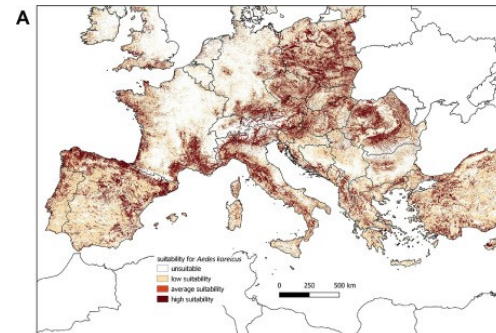
- Malta
- Monaco
- San Marino
- Gibraltar
- Liechtenstein
- Azores (PT)
- Canary Islands (ES)
- Madeira (PT)
- Jan Mayen (NO)



ECDC and EFSA, map produced on 6 Oct 2023. Data presented in this map are collected by the VectorNet project. Maps are validated by external experts prior to publication. Please note that the depicted data do not reflect the official views of the countries.
* Countries/Regions are displayed at different scales to facilitate their visualisation. The boundaries and names shown on this map do not imply official endorsement or acceptance by the European Union. Administrative boundaries © EuroGeographics, UNFAO.

Assessing the distribution of invasive Asian mosquitoes in Northern Italy and modelling the potential spread of *Aedes koreicus* in Europe

Irene Arnoldi ^{a b c d}, Agata Negri ^{a b}, Laura Soresinetti ^{a c}, Mattia Brambilla ^a,
Davide Carraretto ^{c d}, Fabrizio Montarsi ^f, Paolo Roberto ^g, Andrea Mosca ^g, Diego Rubolini ^e,
Claudio Bandi ^{a b}, Sara Epis ^{a b} , Paolo Gabrieli ^{a b}



The potential invasion into North America and Europe by non-native mosquito, *Aedes koreicus* (Diptera: Culicidae)

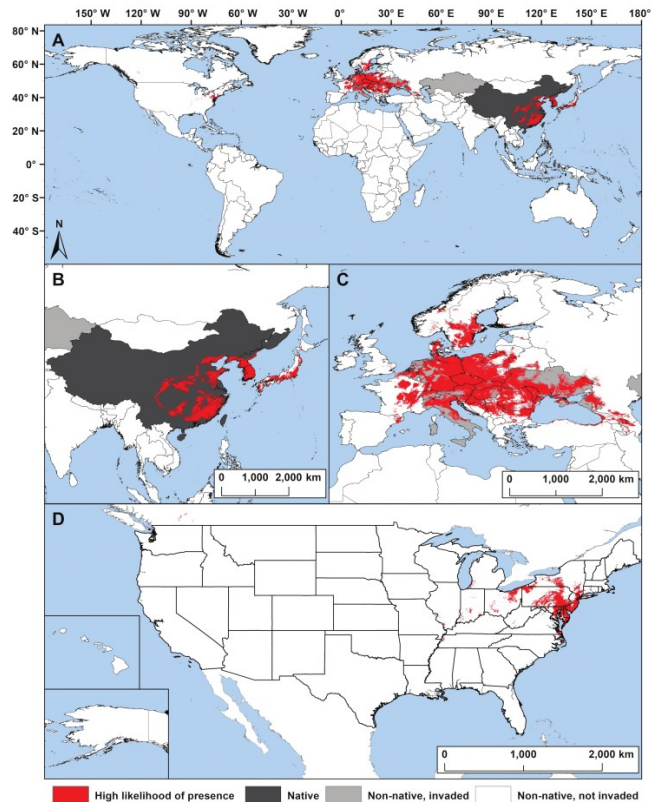
[Get access >](#)

Sangwoo Seok, Zhonghyun Kim, Valerie T Nguyen, Yoosook Lee ✉

Journal of Medical Entomology, Volume 60, Issue 6, November 2023, Pages 1305–1313,

<https://doi.org/10.1093/jme/tjad116>

Published: 31 August 2023 Article history ▾



Potential Global Distribution of the Invasive Mosquito *Aedes koreicus* under a Changing Climate

Qing Liu^{1,†}, Jing-Wen Xie^{1,2,†}, Ming Wang¹, Yu-Tong Du¹, Zi-Ge Yin¹, Ning-Xin Zhou¹, Tong-Yan Zhao¹, En-Jiong Huang^{2,3,*} and Heng-Duan Zhang^{1,*}

¹ State Key Laboratory of Pathogen and Biosecurity, Beijing Institute of Microbiology and Epidemiology, Beijing 100071, China

² The School of Public Health, Fujian Medical University, Fuzhou 350000, China

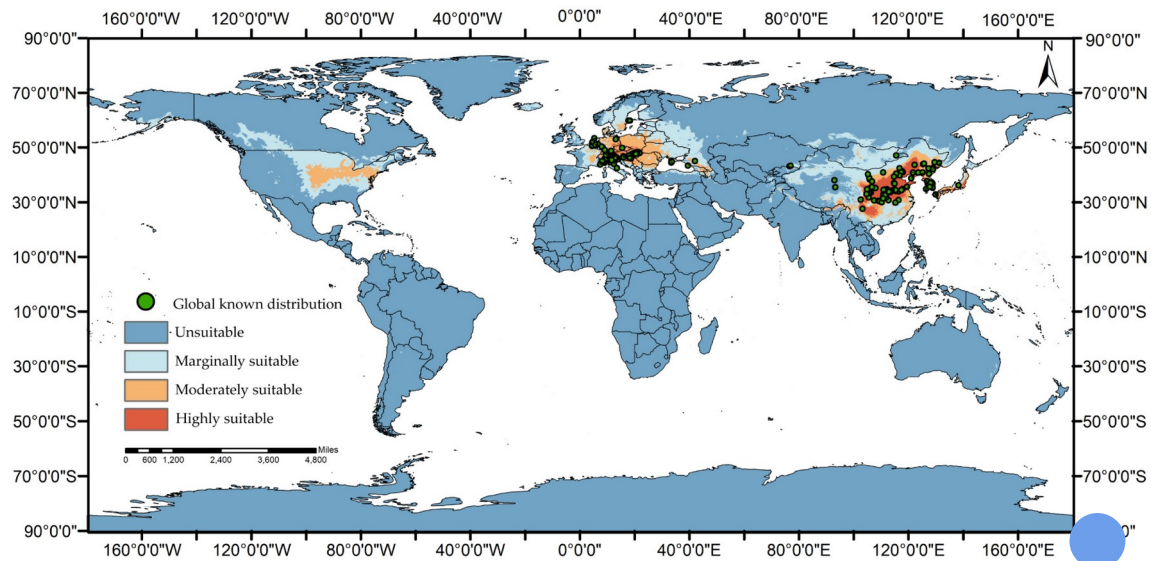
³ Fuzhou International Travel Health Care Center, Fuzhou 350001, China

* Authors to whom correspondence should be addressed.

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Trop. Med. Infect. Dis. **2023**, 8(10), 471; <https://doi.org/10.3390/tropicalmed8100471>

This article belongs to the Section **Infectious Diseases**



What we can do? **Limitations** of correlative models

With **limited** and **opportunistic observations**, this is often the **best we can do**, but:

- Sparse observations
- Our correlative model efforts have mostly used multidecadal climatic averages
- Sampling bias can fool our perception of the species distribution and spread

The Map Is Not The Territory

- Alfred Korzybski

'The *menu* is not the *meal*. – Alan Watts

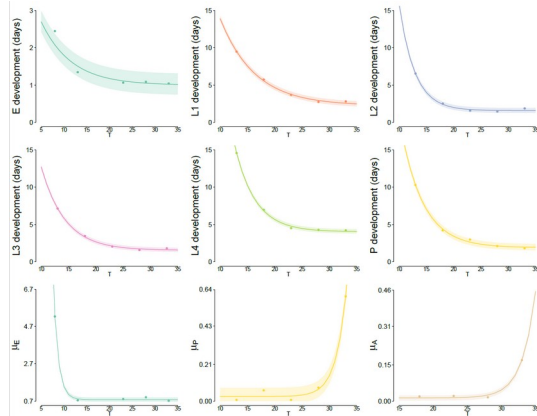


La trahison des images - René Magritte



What we can do? Mechanistic models

Laboratory experiments



$$\begin{aligned}
 E' &= n_E \tau_A A - (\tau_E + \mu_E) E \\
 L_1' &= \tau_E E - \left(\tau_{L_1} + \mu_{L_1} \left(1 + \frac{L}{K} \right) \right) L_1 \\
 L_2' &= \tau_{L_1} L_1 - \left(\tau_{L_2} + \mu_{L_2} \left(1 + \frac{L}{K} \right) \right) L_2 \\
 L_3' &= \tau_{L_2} L_2 - \left(\tau_{L_3} + \mu_{L_3} \left(1 + \frac{L}{K} \right) \right) L_3 \\
 L_4' &= \tau_{L_3} L_3 - \left(\tau_{L_4} + \mu_{L_4} \left(1 + \frac{L}{K} \right) \right) L_4 \\
 P' &= \tau_{L_4} L_4 - (\tau_P + \mu_P) P \\
 A' &= \frac{1}{2} \tau_P P - (\mu_A + \chi \alpha \tau_A) A \\
 C' &= \chi \alpha \tau_A A
 \end{aligned}$$



RESEARCH **Open Access**

First report of the influence of temperature on the bionomics and population dynamics of *Aedes koreicus*, a new invasive alien species in Europe

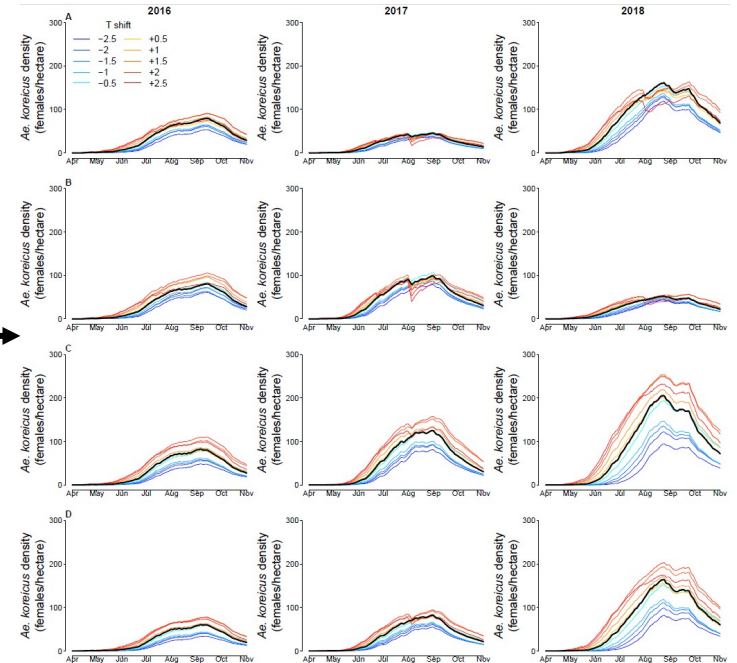
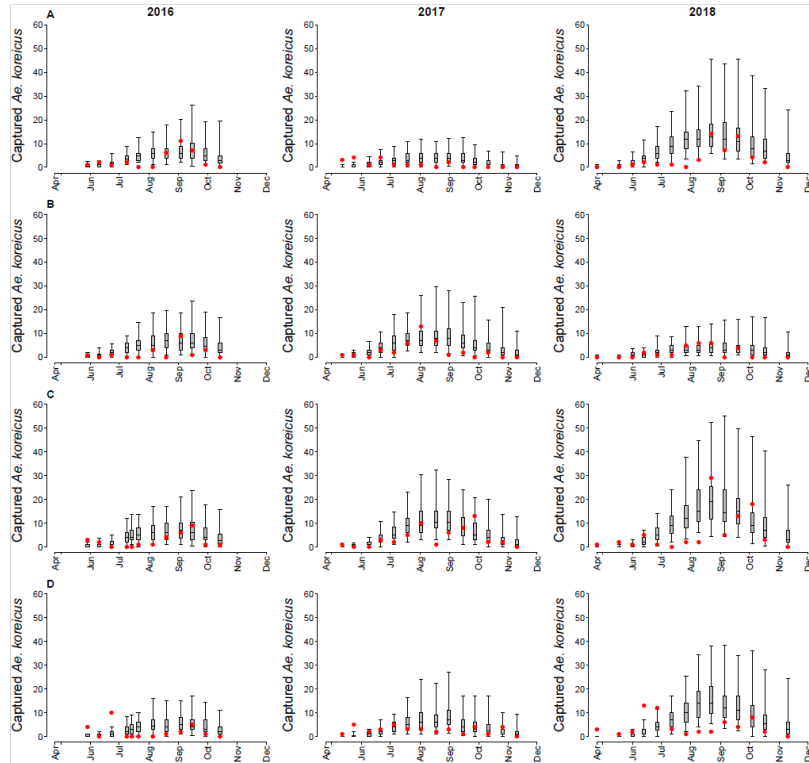
Giovanni Martini^{1,2*}, Daniele Arnoldi³, Federico Baldacchino^{1,3}, Gioia Capelli⁴, Giorgio Guzzetta^{2,3}, Stefano Meier^{1,3}, Fabrizio Montanari², Annachiara Rizzoli² and Roberto Rosa^{1,2}



First report of the influence of temperature on the bionomics and population dynamics of *Aedes koreicus*, a new invasive alien species in Europe

Giovanni Martini^{1,2*}, Daniele Amadi³, Frederic Baldacchino^{1,3}, Gioia Capelli³, Giorgio Guzzetta^{2,3}, Stefano Meier^{1,3}, Fabrizio Montarsi³, Annapaula Rizzoli³ and Roberto Rosa^{1,3*}

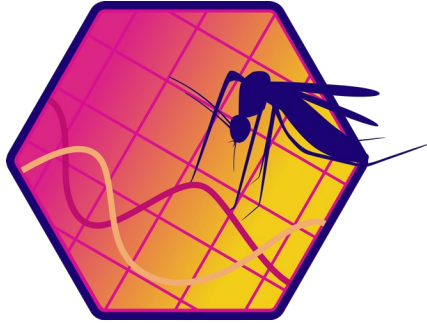
What we can do? Mechanistic models



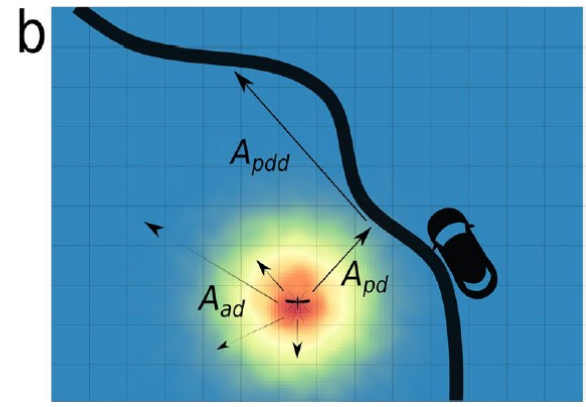
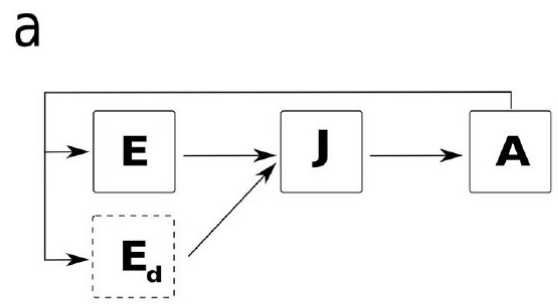
Model fitting and calibration

Alternative temperature scenarios

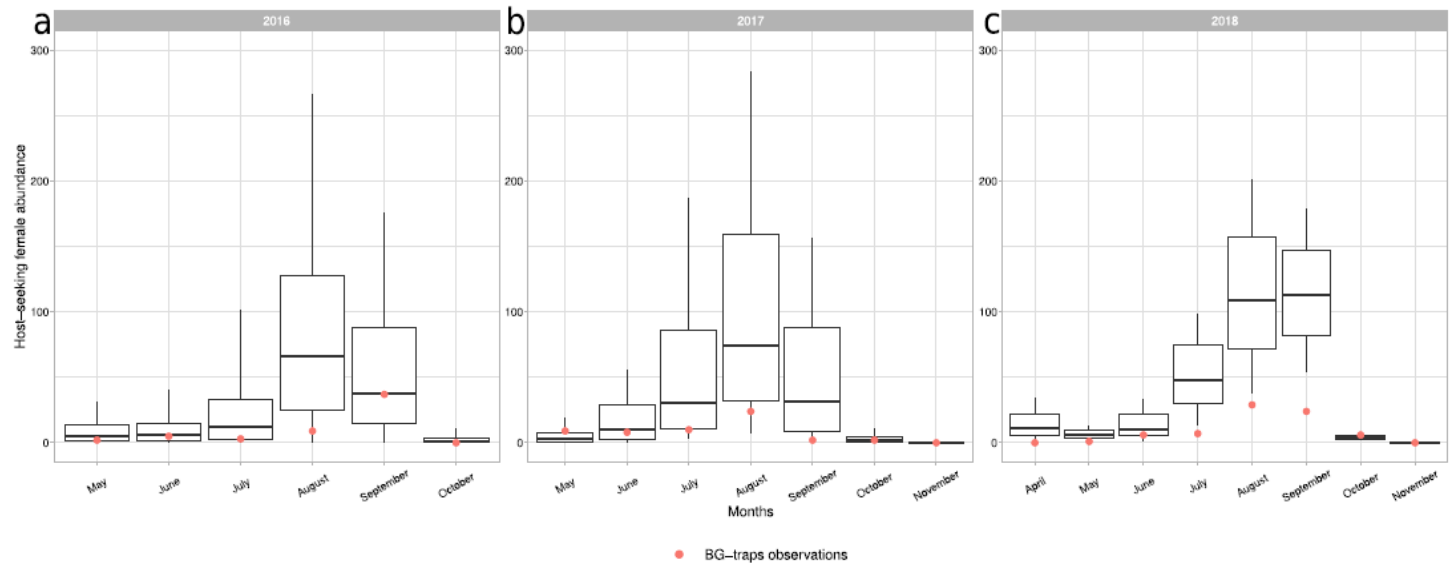
What we can do? Mechanistic models



dynamAedes is R library providing a stochastic, time-discrete and spatially-explicit population dynamical model for four invasive *Aedes* mosquito species: *Aedes aegypti*, *Ae. albopictus*, *Ae. japonicus* and *Ae. koreicus*.



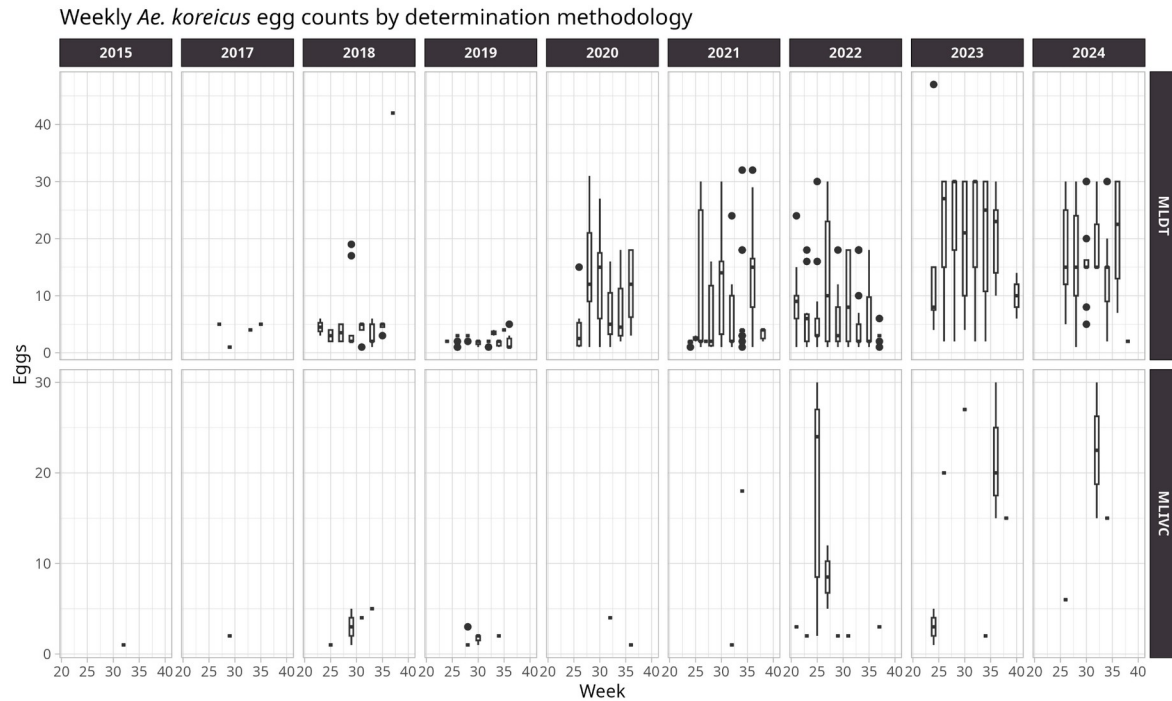
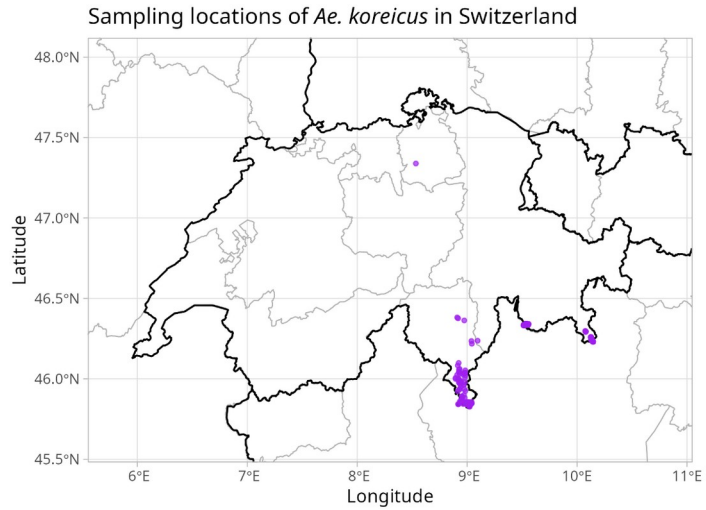
What we can do? Mechanistic models



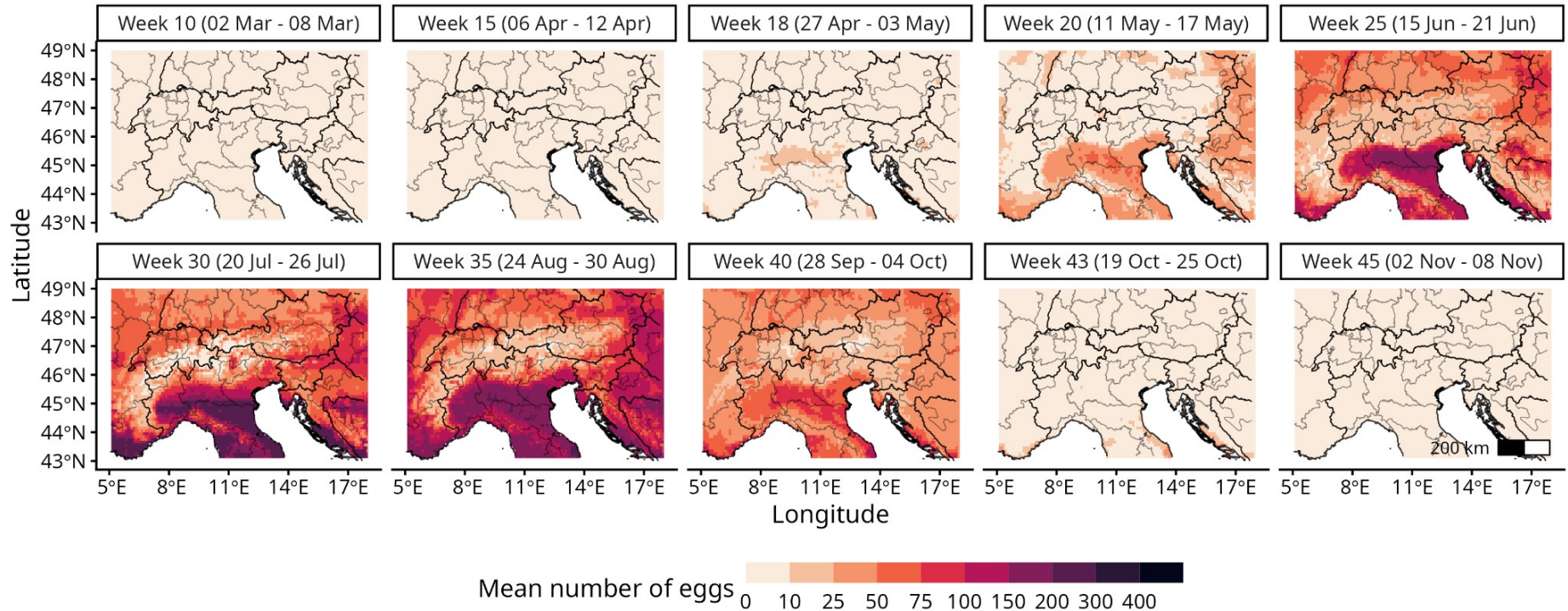
What is missing?

- **The Observational Data Bottleneck:** lack of high quality time series data for *Aedes koreicus* (both ovitrap and adult collections).





Example: *Ae. albopictus* ovipositing activity seasonal forecast



What is missing?

- **The Observational Data Bottleneck:** lack of high quality time series data for *Aedes koreicus* (both ovitrap and adult collections).
- **Biological Knowledge Gaps:** scarcity of empirical data regarding Thermal Performance Curves (TPCs)



What is missing?

- **The Observational Data Bottleneck:** lack of high quality time series data for *Aedes koreicus* (both ovitrap and adult collections).
- **Biological Knowledge Gaps:** scarcity of empirical data regarding Thermal Performance Curves (TPCs)

Data availability, rather than modelling methods or environmental covariates, **remains the primary limiting factor** in our understanding of *Aedes koreicus*.



What is missing?

- **The Observational Data Bottleneck:** lack of high quality time series data for *Aedes koreicus* (both ovitrap and adult collections).
- **Biological Knowledge Gaps:** scarcity of empirical data regarding Thermal Performance Curves (TPCs)

Data availability, rather than modelling methods or environmental covariates, **remains the primary limiting factor** in our understanding of *Aedes koreicus*.

Defining the Research Question: **what do we want to know?**



Some open questions (among the many)

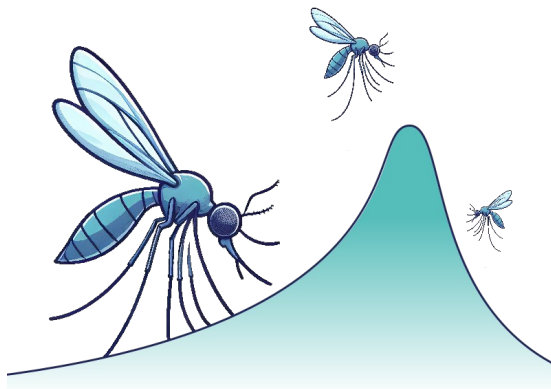
- **Altitudinal Expansion:** To what extent can this species move into higher mountain areas as a direct result of climate warming?
- **Interspecific Competition:** Will *Aedes albopictus* eventually outcompete *Aedes koreicus* in shared niches, or can they coexist through niche partitioning?



Some open questions (among the many)

- **Altitudinal Expansion:** To what extent can this species move into higher mountain areas as a direct result of climate warming?
- **Interspecific Competition:** Will *Aedes albopictus* eventually outcompete *Aedes koreicus* in shared niches, or can they coexist through niche partitioning?
- **Impact Assessment:** Is the primary threat of this species a matter of public health risk, or is it predominantly a social nuisance issue?
- **Trophic Plasticity:** We still lack a clear understanding of its host preferences. Is this species more ecologically plastic than we previously expected?
- **Active and passive dispersal capacity?**





Thank you for your attention

Daniele Da Re
Giovanni Marini



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EDMUND MACH
dal 1874



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